WARM ROOF: PROTEUS PRO-FELT® 'ULTIMA-PLUS' (BUR)

ONTO CONCRETE ROOF DECKS

PROTEUS SYSTEM AND MEMBRANE RANGE

PROTEUS PRO-FELT® HIGH PERFORMANCE MEMBRANES
PROTEUS PRO-SYSTEM® COLD INSTALLED, SEAMLESS WATERPROOFING
PROTEUS PRO-THERM PIR WARM ROOF INSULATION
COLD MELT® BURIED/GREEN WATERPROOFING
PROTEUS HOT MELT® INVERTED/GREEN/ PODIUM DECK SYSTEMS
PROTEUS PRO-THERM XPS INVERTED ROOF INSULATION
PRO-BW® PLUS FAST CURE BALCONY WATERPROOFING

NFRC LEADING ROOFING EXCELLENCE
LRWA Liquid Roofing and Waterproofing Association
CONFEDERATION OF ROOFING CONTRACTORS
Safe2 Torch
BBA APPROVAL INSPECTION TESTING CERTIFICATION CERTIFICATE 17/5432

S_concrete plywood_PFB20_170323

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Project Number: **Project Name:** **Document Reference:**	2221717 Tradewinds, Wards Wharf Approach, London E16 2EY Main Roof Block A,B & C S_ 2221717_Tradewinds_PF
Prepared for:	Urang Property Management 196 Kings Rd London SW6 4NF Tom Theakstone tom.theakstone@urang.co.uk 020 4538 5700 Date: 27th April 2023
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INTERNAL USE ONLY	1200 Concrete

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Description:	**PROTEUS PRO-FELT® ULTIMA 'PLUS'** is a plastomeric, high quality modified bitumen torch-on membrane that excels as a direct waterproofing overlay to existing substrate or as a system element of the robust **PROTEUS WATERPROOFING** built-up roof system. This Specification guides the user through the correct installation method of the **PROTEUS WATERPROOFING** high-performance **PROTEUS PRO-FELT® ULTIMA 'PLUS'** Built-Up Roof (BUR) onto existing concrete, existing or new timber / WBP plywood roof decks in conjunction with **PROTEUS PRO-THERM INSULATION**. This document is to be read in conjunction with the separate **PROTEUS WATERPROOFING** Roof Condition Report. Proteus accept no liability for any defects arising where the existing roof condition is not fully checked and confirmed by the contractor prior to commencement of works.

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SPECIFICATION OUTLINE

- Remove the existing roof coverings to expose the roof deck
- Prepare, repair and clean surfaces and prime surfaces as required
- Install **PROTEUS PRO-FELT®** air vapour control
- Bond **PROTEUS PRO-THERM** insulation bonded with **PRO-BOND FOAMING**
- Install **PROTEUS PRO-FELT®** underlays
- Install **PROTEUS PRO-FELT®** mineral cap-sheet

Refer to the Product Schedule and main Specification for full product details and usage, coverage rates and non-combustible insulation placement.

PRIMERS	
Pro-Prime® SA	Primer for Proteus self-adhesive membranes
Pro-Prime® Bitumen	Primer for Proteus torch-on membranes Note: Torch receivable Proteus Pro-Therm Insulation boards are pre-primed
SELF-ADHESIVE AIR VAPOUR CONTROL / ADHESIVE / INSULATION	
Proteus Pro-Felt® SA Underlay Sanded	Self-adhesive air vapour control
Pro-Bond Foaming	Adhesive for insulation boards
Proteus Pro-Therm Insulation CTF	Tapered insulation scheme
UNDERLAYS / CAPSHEET	
Proteus Pro-Felt® Ultima 'Plus' Underlay	Torch on underlay – horizontal
proteus pro-Felt® 3mm underlay plain	Torch-on underlay – verticals, details
Proteus Pro-Felt® Ultima 'Plus' Mineral Capsheet	High performance torch-on capsheet

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SAFE2TORCH MEMBRANES / ANCILLARIES:

SAFE2TORCH GUIDELINES MEMBRANES	
Proteus Pro-Felt® SA Underlay Sanded	Self-adhesive air vapour control/underlay
Proteus Pro-Felt® SA Mineral Capsheet	Self-adhesive mineral capsheet
ANCILLARIES	
Proteus Edge Trims	Edge Trims
Proteus Bitumen Sealant	Sealant

For ancillaries, roof-lights and roof hatches contact **PROTEUS WATERPROOFING** on:
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CODES OF PRACTICE / BUILDING REGULATIONS:

Building Regulations:
Building Regulations Part L: Approved Document L1B Conservation of fuel and power in existing buildings requires that (when re-roofing), the existing roof construction must achieve the threshold U-value of **0.35 W/m²K** or better. If the threshold value is not achieved, then the roof must be thermally upgraded to meet the current required maximum U-value of **0.16 W/m²K**.

Building Regulations Part B: Regulations for fire safety in residential homes, including new and existing dwellings, flats, residential accommodation, schools, colleges, and offices. For these regulations to be correctly met, further information may be required to establish the appropriate insulation choice for this Specification; particularly above internal party walls and the provision of non-combustible insulation where necessary.

This Installation Specification does not automatically meet **Part B** as further information about the general construction is required from the client to establish the requirement regarding compartmentation to confirm **Building Regulations Part B** compliance.

It is important to note that it is impossible to test all possible roof build-ups against **Broof(t4)**, as there are too many variables. Relying solely on advice by proxy may not provide an accurate assessment of the roof's fire performance which could potentially lead to safety issues in the event of a fire. It is therefore the responsibility of the client to instruct the project building control officer, or an independent fire engineer to review the proposed Proteus specification, request any further information that may be required, and confirm suitability and satisfaction of the relevant national building regulations, prior to the commencement of works.

For thermal calculations (**Part L**) non-combustible insulation compliance (**Part B**) and CTF tapered insulation schemes contact **PROTEUS WATERPROOFING** on:
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PRELIMINARIES / HEALTH & SAFETY / TORCH ON GUIDELINES:

For this project, it is the Clients responsibility to manage the project and contract and ensure suitable arrangements are in place to conform to the **Construction (Design and Management) Regulations 2015 Regulation 4**.
PROTEUS WATERPROOFING will assist with the design of the flat roof / balcony system though not undertake the role of Principal Contractor. Where the project requires more than one Contractor it is the Client responsibility to appoint a Principal Designer and a Principal Contractor to conform to the **Construction (Design and Management) Regulations 2015 Regulation 5**.

Structure and Loadings:
PROTEUS WATERPROOFING does not advise on structural loadings and cannot confirm capacity or limitations in regard to the suitability of the structure to take the weight of the proposed system, or any finishes such as decking, paving, gravel, living roofs, solar panels, plant and equipment or any other items situated on the roof, balcony, or terrace. It is the responsibility of the client to take separate advice from an independent structural engineer to ensure that the structure is suitable for the final finished weight and overall loadings placed onto the roof, balcony, or terrace.

Additional Loadings:
This roof, balcony or terrace specification forms a change from the existing roof build-up and has been designed to conform with **BS 6229** as agreed with the client. It is the responsibility of the client to take separate advice from an independent structural engineer to ensure that the structure is suitable for the final finished weight and overall loadings placed onto the roof, balcony, or terrace.

Before tendering, the contractor should examine the drawings and specification documents, visit the site, and ascertain all local conditions and restrictions, accessibility, the full extent and nature of the work, the supply and conditions affecting labour and the execution of the contract generally. No claims arising from failure to do so will be considered.

The contractor is to take their own roof core samples to satisfy themselves regarding the existing roof build-up and to ascertain the extent of the work required in removing the existing roof covering. No claims arising from failure to do so will be considered.

All roofing materials are to be supplied by **PROTEUS WATERPROOFING** and to be fit for purpose and of the type and quality described herein. Any sub-standard materials will be rejected. No alternatives are to be used without written consent by **PROTEUS WATERPROOFING**.

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All products must be stored according to the Manufacturer instructions, ideally inside a building or site container. However, if external storage is unavoidable, store clear of the ground or supporting surface and protected at all times from moisture, frost, and mechanical damage. Rolls of waterproofing to be stored on end on a flat firm surface. Cover with a waterproof sheet allowing venting to avoid condensation build up.

The roof waterproofing work should adhere to **Building Regulations current document, Part L**. It is recommended confirmation of individual project requirements for thermal performance of the roof is sought by the client or the client's representatives from the local Building Control office before work commencement. When installing multi-layer waterproofing membranes without insulation, **PROTEUS WATERPROOFING** understands that the consideration to thermal performance compliance under **Building Regulations Part L** is being dealt with separately by the client, and as requested will not form part of our recommendations.

Please contact **PROTEUS WATERPROOFING** to ascertain the correct **U Value** to be achieved.

It is the responsibility of the roofing contractor to liaise with the Main Contractor or Client concerning the possible temporary disruption or suspension of existing services to the building during the works.

Specified Exceptions:
PROTEUS WATERPROOFING specifically excludes the following, and cannot be held responsible or liable for any defects arising from;
Structural failures caused by additional loadings onto the roof, subsidence, heave or landslip to the building, fire, explosion, lightning, earthquake, storm, flood, bursting and overflowing of water tanks, apparatus or pipes, impact by aircraft and articles dropped from them, impact by vehicles, riot, civil commotion, malicious or deliberate damage, any other accidental damage, other third party installations, third party expansion joints, other third party materials including EPDM door/window seals, third party construction details not meeting relevant Codes of Practice including **BS 6229**, inherent or latent defects of the building or existing roofing system including the roof deck or the substrate, substrate movement, new or existing interstitial condensation, discharge or spillage of any third-party substance (chemical or otherwise), water entry from adjacent structures or anywhere else in the building other than from the finished system, lack of maintenance and abnormal use.

The contractor shall employ none but fully qualified, competent tradesmen and the whole of the work shall be carried out and completed in accordance with industry "Best Practice".

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The contractor shall carry out the works without undue inconvenience and nuisance and without danger to occupants and users.

Following completion of the installed, specified roofing works, provision should be given to the protection of all finished surfaces from damage by subsequent building operations. Additional work will be required to return the membrane to a warrantable condition if the membrane is damaged due to inadequate protection.

Any elements that may affect the future performance of the specified system shall be reported to the client and **PROTEUS WATERPROOFING** at the earliest possible opportunity. It is the responsibility of the contractor to acquaint themselves with all relevant codes of practice referred to within the specification. **PROTEUS WATERPROOFING** takes no responsibility for misinterpretation or lack of knowledge for third parties.

The works shall be carried out in accordance with the requirements of **BS 6229** Code of Practice for flat roofs with continuously supported coverings and **BS 8217** Reinforced bitumen membranes for roofing Code of Practice.

The roofing membrane, insulation, and any ancillary items, (roof lights, trim, outlets) included in this specification shall be obtained from:

PROTEUS WATERPROOFING
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Tel: 01268 777871

The works shall be carried out in accordance with all current relevant legislation, and recommendations from the **Health & Safety Executive (HSE)** including **HSG33 Health & Safety** in roof work.

The Contractor is to comply with **Control of Substances Hazardous to Health Regulations 1988**. All relevant information relating to hazard assessment to be submitted with tender.
The Contractor is to allow for all necessary temporary protection to leave the site weather tight at all times.

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The Contractor is to allow for the removal from the site of all debris and surplus materials, which have arisen from the execution of the works.

The Contractor is to allow for protection of the works from traffic and damage during the construction period.

Where possible any newly installed roof decking should be protected and covered from the weather to ensure drying out is not necessary. However, in the event of a roof becoming wet, it is important to ensure drying out is done safely.

There are several methods of drying out a roof that can be employed by the roofing contractor, but the most common method used due to speed and effectiveness is still by use of a gas torch. When a gas torch is to be used for drying, a thorough roof inspection of all fire related risks is essential. Where risks are identified, the operative must adopt a torch free area in line with the contractor's public liability insurance, however, it is recommended that this should be no less than **900mm**.

Methods of drying within a restricted area will vary from wiping with dry rags, and mops, then allowing natural weather conditions to finish the drying, or the use of hot air guns.

Please note that it is your responsibility as the contractor to ensure that all the recommendations detailed in this specification, and any changes or additions subsequently made, with or without Proteus' knowledge, particularly in respect of insulation design, are approved by a building control fire safety officer prior to the commencement of any works on site.

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TORCH ON GUIDELINES

It is the responsibility of the contactor to perform a thorough roof inspection of all fire-related risks and adopt a torch free area no less than **900mm** from all 'at risk' areas. At risk areas are to include, but are not limited to, the following:

- Timber upstands
- Hanging tiles
- Thatched roof
- Roof-light Kerbs and upstands
- Cladding
- Lantern roof lights
- Confined space
- Window cill

Thought should be given to junctions to existing waterproofing with flammable insulation or substrate materials. These areas and existing weathering components need to be assessed and made Safe2Torch before proceeding with hot works, alternatively a torch-free solution should be considered. Attention needs to be given to concealed flammable materials where there is a chance the flame could travel and ignite particles in inaccessible areas, these include:

- Louvered vents, air ducts, intakes, and outtakes
- Timber, DPC or sarking membranes beneath fixed metal capping system's
- Existing kitchen extraction plant coated in oils or fats
- Flammable wrapping to trunking/ducting
- Existing metal or plastic coping/capping's
- Existing valuable plastic curbs, domes, pipes

When working near recently applied bituminous primer or other solvent-based coatings, torching needs to be avoided.

Other pre-work checks should include an assessment of the substrate - under no circumstances should a torch be applied direct to a timber roof deck or timber upstands, including timber fillets, even if the substrate has been treated with bituminous primer. A self-adhesive base-layer or a mechanically fastened layer should be used to protect the timber substrate coming into contact with the naked flame. A gas torch should never be used directly on a timber roof deck.

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Expansion joints which are filled with combustible fillers, such as foam or fiberboard, should not come into contact with high heat or naked flame.

If a membrane roof is adjoining a pitched roof with details to be completed under abutments to roof tiles and slates, the slates or tiles should be removed, and a compatible torch-free product should be selected.

A compatible torch-free product should be selected for any abutments with open cavities or open perpends.

Naked flames should be avoided where there is fixed timber, plastic fascia's or soffits.

If further information is required, the full Safe2Torch guidelines can be found at https://www.nfrc.co.uk/docs/default-source/campaign/safe2torch-guidance.pdf?sfvrsn=2

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PRE-WORKS

Installation Specification Validity: This Installation Specification is valid for a period of **12 months**, after which time it is recommended a further inspection is carried out in case changes require amendments to the proposed Specification. Should a site inspection by **PROTEUS WATERPROOFING** not have been carried out, this is to be arranged prior any works commencement to confirm the Specification. The site inspection may require changes and amendments to the proposed Specification. Any measurements stated in this document are approximate and not to be used for tendering. The tender document or Client is to be consulted to confirm the areas of work to be quoted. The Contractor is to carry out their own investigations, core sampling, and confirming detailing, to confirm this Specification is suitable.
Structure and Loadings: **PROTEUS WATERPROOFING** does not advise on structural loadings and cannot confirm capacity or limitations in regard to the suitability of the structure to take the weight of the proposed system, or any finishes such as decking, paving, gravel, living roofs, solar panels, plant and equipment or any other items situated on the roof, balcony, or terrace. It is the responsibility of the client to take separate advice from an independent structural engineer to ensure that the structure is suitable for the final finished weight and overall loadings placed onto the roof, balcony, or terrace.
Additional Loadings: This roof, balcony or terrace specification forms a change from the existing roof build-up and has been designed to conform with **BS 6229 / National Building Regulations** as agreed with the client. It is the responsibility of the client to take separate advice from an independent structural engineer to ensure that the structure is suitable for the final finished weight and overall loadings placed onto the roof, balcony, or terrace.
Substrate: This Document refers to the installation of **PROTEUS PRO-FELT® 'ULTIMA-PLUS' BUR** onto existing concrete, existing and new timber / WBP plywood roof decks. Prior to the installation of the new roof system the Client is to check the roof deck maximum weight loadings capacity and ensure the roof deck can accept the new/increased weight loadings. A structural engineer may be required to carry out the necessary calculations. Substrates must be without negative deflections or backfalls.
Removing the Existing Roof Coverings: The existing roof covering are to be carefully removed and disposed of, exposing the existing roof deck.
Roof Decks: Roof decks require a minimum finished fall of **1:80** unless **PROTEUS PRO-THERM® INSULATION CTF** tapered insulation schemes are installed in which case roof decks are required to be level.

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Insulation Schemes including CTF: When installing a **PROTEUS PRO-THERM FLAT BOARD INSULATION** scheme minimum deck falls of **1:80** is assumed. When installing a **PROTEUS CUT TO FALLS (CTF) INSULATION** scheme a flat deck is assumed. Obstructions such as roof lights, original deck deflections and other checks may create some areas of standing water. Standing/frozen water will not adversely affect the performance of **PROTEUS PRO-THERM INSULATION. PROTEUS CTF INSULATION** schemes are intended to provide adequate drainage off the roof should a flat board provide insufficient run off.
Deflections: Existing deflections will not be resolved with a waterproofing only or including a flat board insulation overlay. Ensure these are levelled prior the waterproofing overlay works proceeding.
Roof Plant: If M&E, Air Conditioning Units, Extractor Fans, and Cables/Trays cannot be temporarily moved or raised and therefore impede access for the installation of the new roof system, these areas may be excluded from the waterproofing guarantee, or a guarantee may not be possible.
Aerials, Satellite Dishes, Telecommunications: The Contractor is to establish if equipment emitting harmful radiation is present on the roof. Where these impede works, temporarily remove to allow access.
Lightning Conductors: Qualified personnel to temporarily divert and adapt as required to allow for the roofing works. Reinstate/renew lightning conductors on completion of the roofing works.
General Building Works: Any associated building works, such as (But not limited to) cutting chases, brick-work repairs, mortar joint repairs and rendering should be carried out prior to the installation of the waterproofing membrane.
Up-Stands and Windows: All upstands must conform to **BS 6229** and to be a **Minimum 150mm** high from the finished roof level. **Note:** The finished roof level is the top surface of the waterproofing unless there are roof finishes. I.e., Ballast, Decking Systems, Paving Slabs or Green Roofs. In which case, the finished roof level is to be **Minimum 150mm** above the surface finishes.
New WBP Plywood to voids or parapets: Install a new **minimum 18mm** WBP Plywood board conforming to **BS EN 1995** and **BS 8217.** Fix the new WBP Plywood directly to the rafters with appropriate length non corroding screw fastenings. Tape the WBP Plywood ready to receive the new waterproofing membrane.
Adapting Parapet Walls for Metal/Lead Capping: Temporarily remove the capping to allow the installation of **PROTEUS PRO-FELT®** membrane. Adapt the parapet as required and reinstate/renew the capping.

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Adapting Parapet Walls: **18mm** WBP Plywood conforming to **BS EN 1995** and **BS 8217.** Fix the new WBP Plywood directly to the rafters with appropriate length non corroding screw fastenings. Prime the WBP Plywood ready to receive the new waterproofing membrane to a new **PROTEUS EDGE TRIM** at the external perimeter. To mitigate the risk of cold bridging inside voids, the contractor must ensure all voids between the back of the new WBP plywood, and the parapet wall are in-filled with appropriate insulation at a minimum of **30mm** thick and **300m** high on the verticals from the roof surfaces.
Retained Internal Outlets, Parapet Outlets: Inspect all down pipes and obtain confirmation their capacity is appropriate for the roof area, to ensure free flow of wastewater. Where the existing outlets / downpipes are to be retained, inspect for any damage and repair as required. It is recommended ground drainage and drainage pipes are inspected using CCTV equipment to ensure these are free of damage and allow a free flow of wastewater.
Detail Integrity: Check integrity of the existing substrate, skirtings, and collars. Other details should be cut back if required, making sure the substrate is sound and dimensionally stable. All protrusion pipes should be extended where necessary to accommodate new waterproofing level.
Edge Details: Existing defective trims, lead flashings and up-stand details must be removed and renewed ready to receive the new items.
Overflows: Raise these so the exit pipe is above the new skirting height, to be a minimum of **150mm** high from the finished waterproofing level.
Damp Proof Courses, Cavity Trays: Do not install the new waterproofing above damp-proof courses or cavity trays; raise these where required. Details which subsequently do not meet the required **150mm** height must be discussed with **PROTEUS WATERPROOFING**.
Roof Openings: At the end of each day ensure all openings are sealed to prevent water ingress. Identify redundant penetrations, which should be removed and disposed of. Any openings are to be sealed to prevent water ingress.
Defective Timber/uPVC: Remove and replace any defective timber to the existing roof such as timber decks, structural joists, timber fillets, barge boards, and fascia's hard edges.
Repair Tape: ALL self-adhesive 'flashing' tape which has previously been used as a means of repair must be completely removed. **Note: Substrate defects may well be hidden by such tapes.**
Angle Fillets: Install a suitable non-combustible angle fillet to all junctions at horizontal to vertical without exception.

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Perimeter Check Curb: Raise the curbs to a minimum of **50mm x 50mm** above the drip battens.

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SURFACE PREPARATION: CONCRETE ROOF DECK

For Concrete, Screeded Roof Decks: All cracks, gaps, and holes should be raked out, suitably prepared, and repaired. Hollows, blows, or ridges in the existing substrate are to be levelled to leave a wood float finish. Any cracks above **5mm** should be repaired with **PROTEUS FASTFILL** rapid setting mortar. The roof deck must be dry, free of laitance and sound and have a minimum finished fall of **1:80** unless **PROTEUS PRO-THERM INSULATION CTF** tapered insulation schemes are installed in which case roof decks are required to be level.
Clear/Prepare Surfaces: Remove, scrape, and sweep away the bulk of contamination. Do not block drainage system with debris. All substrates must be dry before applying the membrane.
Existing Surfaces: The existing substrate must be sound, dimensionally stable, clean, and free of loose matter and excess water. The existing surface may be used as an air vapour control layer (AVCL) if a surface inspection for defects such as delamination blisters, failed lap-joints and ridging demonstrates sound integrity. A recovery layer can be introduced should the substrate be beyond repair. An AVCL must be introduced to concrete and timber decks. All surfaces are to be level ready to accept the specified system.

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Registered Office: Charter House, 105 Leigh Road, Leigh-On-Sea, Essex, SS9 1JL | Registered company number: 08458402

PRODUCTS: PRIMING

Proteus Pro-felt® Torch-on Membranes: To surfaces to receive torch-on membranes apply **PRO-PRIME® BITUMEN** @ a minimum rate of **0.20ltr/m²**.
Proteus Pro-felt® Self-adhesive Membranes: To surfaces to receive self-adhesive membranes apply **PRO-PRIME® SA** @ a minimum rate of **0.20ltr/m²**.
Note: PROTEUS PRO-THERM INSULATION boards do not require to be primed to receive **PROTEUS PRO-FELT® UNDERLAYS.**

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PRODUCTS: SELF-ADHESIVE MEMBRANES / TORCH-ON MEMBRANES

Self-adhesive Membranes Safe2torch for Fire Risk Areas:
PROTEUS PRO-FELT® torch-on air vapour controls, underlays and mineral cap-sheets are technically suitable to be used, where the contractor deems it safe to do so. Where there is a risk of combustion **PROTEUS PRO-FELT® SA** self-adhesive membranes are to be used to remove the use of open flame. Where the membrane is to be installed directly to combustible substrates such as WBP plywood, timber, or near combustible materials such as insulation behind cladding, up-stands under pitched tiled roofs. Then torch-on membranes must not be used.

Primer:
PROTEUS PRO-FELT® SA membranes require all surfaces to be first primed with **PRO-PRIME SA.**
Air Vapour Control and Underlay:
In fire risk areas install the self-adhesive **PROTEUS PRO-FELT® SA UNDERLAY SANDED**.
Cap-sheet:
In fire risk areas install the self-adhesive **PROTEUS PRO-FELT® SA MINERAL CAP-SHEET.**

Note: For further clarification refer to the **NFRC 'Safe2Torch' Guidelines**.

Self-adhesive Membranes Side Laps, and End Laps:
PROTEUS PRO-FELT® SA membrane is to be installed onto the primed substrate or already in-situ **PROTEUS PRO-FELT®** air vapour control/underlay/capsheet up to the termination on the up-stand and fully seal by continuing by a minimum of **200mm** onto the horizontal.

All end and side laps are **HOT-AIR** activated to produce a homogeneous bond and produce a bitumen extruded bead from the lap. Side Laps are to be a minimum of **80mm,** and head laps are to be a minimum of **100mm** to ensure a watertight seal.

Torch-on Membranes Side Laps, and End Laps:
PROTEUS PRO-FELT® TORCH-ON MEMBRANESA side laps and end laps are to be fully torched on and sealed. Side Laps are to be a minimum of **80mm,** and head laps are to be a minimum of **100mm** to ensure a watertight seal.

Torch-on and **Self-adhesive Membranes Bitumen Bleed:**
It is a guarantee requirement with all **PROTEUS PRO-FELT®** torch-on and self-adhesive membranes to ensure a minimum **5mm-10mm** bitumen bleed is achieved.

Upstands, Skirtings Higher Than 600mm: Underlay
Where the upstands, and skirtings are higher than **600mm**, further mechanical fasteners are required. The fasteners are to be suitable and include a pressure plate.
Insulated upstands will require suitable tube fasteners (cold bridging).
The **PROTEUS PRO-FELT® UNDERLAY** is to be mechanically fixed with fasteners at **300mm** centres.

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Upstands, Skirtings Higher Than 300mm: Torch-on Mineral Capsheet Where the upstands and skirtings are higher than **300mm**, further mechanical fasteners are required. The fasteners are to be suitable and include a pressure plate. Insulated upstands will require suitable tube fasteners (cold bridging). The top leading edge of the **PROTEUS PRO-FELT® TORCH-ON MINERAL CAPSHEET** is to be mechanically fixed with fasteners at **300mm** centres. Once in place, a separate **PROTEUS PRO-FELT® MINERAL CAPSHEET** is to be installed over the fasteners.
Upstands, Skirtings Higher Than 500mm: Self-adhesive Mineral Capsheet Where the upstands and skirtings are higher than **500mm**, further mechanical fasteners are required. The fasteners are to be suitable and include a pressure plate. Insulated upstands will require suitable tube fasteners (cold bridging). The top leading edge of the **PROTEUS PRO-FELT® SELF-ADHESIVE MINERAL CAPSHEET** is to be mechanically fixed with fasteners at **200mm** centres. Once in place, a separate **PROTEUS PRO-FELT® MINERAL CAPSHEET** is to be installed over the fasteners.
Pitched Roofs: Ridge, Hip Capping A detail capping is to be formed separately with the **PROTEUS PRO-FELT® MINERAL CAPSHEET** along the length of the ridge and hip, fully bonded, equidistant and a minimum **150mm** on each pitch.

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PRODUCTS: AIR VAPOUR CONTROLS / INSULATION / WATERPROOFING

The air vapour control is to be continued onto the full skirtings and edge details and must be folded onto the top of the insulation board and sealed compliant with **BS 5250.**
Horizontals Torch on Air Vapour Control: Concrete Surfaces Install by torch on method a layer of **PROTEUS PRO-FELT® 'ULTIMA PLUS' UNDERLAY SANDED** directly to the primed roof areas. Apply the roof torch on the membrane underside.
Verticals Torch on Air Vapour Control: Concrete Surfaces By torch on method install a layer of **PROTEUS PRO-FELT® 3mm PLAIN UNDERLAY** directly to the primed full skirtings, and edge details by a minimum of **200mm** onto the horizontal air vapour control. Apply the roof torch on the membrane underside.
Tapered Insulation Attachment: **PROTEUS PRO-THERM INSULATION CTF** (refer to Proteus tapered insulation scheme). Immediately place **PROTEUS PRO-THERM INSULATION** boards into foamed **PRO-BOND FOAMING** applied @ **50ml/m²** (field application) @ **100ml/m²** (perimeter zones) (see separate Data Sheet). Do not slide boards into place. Do not allow the adhesive to 'skin' before installing insulation boards. (Stagger boards in a brick board format). Sumps to outlets and chutes must be a minimum of **500mm x 500mm** square, and **50mm** lower than the main insulated area. Hard Edges are to be fixed around the sump. **NOTE:** Undulating, rough surfaces may require an expanding adhesive for a secure bond. Double stacked boards to be bonded together with **PRO-BOND FOAMING.**
Hard Edges: Hard edge details must be installed to form the edge against the insulation boards where this is susceptible to damage, when creating a change of direction to the waterproofing including forming internal gutter detail. **Fire Compartmentation:** Galvanised steel angles must be used instead of timber / combustible hard-edge materials along the full party walls line and along the non-combustible insulation return away from the party walls.
Night/Day Joints: Extend **PROTEUS PRO-FELT® SA UNDERLAY SANDED** from the top of **PROTEUS PRO-THERM INSULATION** onto the roof substrate and all details or skirtings by a minimum of **100mm** to form a watertight seal.
Horizontals Torch on Underlay: By torch on method install a layer of **PROTEUS PRO-FELT® 'ULTIMA-PLUS' UNDERLAY** directly to the **PROTEUS PRO-THERM INSULATION** boards.
Verticals Torch on Underlay: By torch on method install a layer of **PROTEUS PRO-FELT® 3mm PLAIN UNDERLAY** directly to the primed full skirtings, and edge details by a minimum of **200mm** onto the horizontal underlay.
Horizontals Torch on Mineral Capsheet: Install by torch on method a layer of **PROTEUS PRO-FELT® 'ULTIMA-PLUS' MINERAL CAPSHEET** onto the already installed underlay.

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Verticals Torch On Mineral Capsheet:
On the skirting install by torch on method a layer of **PROTEUS PRO-FELT® 'ULTIMA-PLUS' MINERAL CAPSHEET** onto the underlay up to the termination detail on the up-stand. Fully seal by continuing the vertical mineral capsheet by a minimum of **200mm** onto the horizontal mineral capsheet.

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PROTEUS PRO-FELT® STANDARD DETAILING

For detail drawings please contact **PROTEUS WATERPROOFING** on: **T: 01268 777871** **E:** sales@proteuswaterproofing.co.uk
Flashings: Dress the **PROTEUS PRO-FELT®** under the new flashing.
Up-Stands, Skirtings, Cills: Dress the waterproofing tightly under the cills/termination detail. Seal with **PROTEUS SEALANT**. All up-stand must conform to **BS 6229** and be a minimum **150mm** high from the finished roof level. **Note:** The finished roof level is the top surface of the waterproofing unless there are roof finishes, i.e., ballast, decking systems, paving slabs, and green roofs, in which case the finished roof level is to be the surface of the finishes.
Perimeter Edge Detail: Fix a new **PROTEUS EDGE TRIM** of an appropriate depth to the perimeter edge. The trims are to be mechanically fixed with screws at **300mm** centres. A bridging piece is to be installed at each section joint. The trim is to overlap any openings by a minimum of **25mm**. The **PROTEUS PRO-FELT®** is to be terminated tightly into the top lip of the primed trim and sealed with **PROTEUS BITUMEN SEALANT.** Where required install new fascia's (uPVC) to the client's specification.
Rainwater Outlets: Ensure the new waterproofing membrane is fully sealed to the outlets and ensure the top flange is situated on top of the underlay. Install a leaf guard. Beneath all rainwater down-pipes, torch on a **PROTEUS PRO-FELT® MINERAL CAPSHEET** piece, **300mm x 300mm**.
Lead Chutes: New Code 5 Lead, form a full height lining to the sides of the opening. Allow a **25mm** turn in under the bridging. Return Lead onto outer wall face with step flashing, dressing into chases. Fully bond both sides of the primed flange between the underlay or soaker sheet and capsheet. Both layers are to be lapped onto the Lead by a minimum of **150mm**.
Pipes and Protrusions: Install the **PROTEUS PRO-FELT®** membrane onto the prepared and primed protrusions **150mm** above the finished waterproofing level. Sleeve the pipe with new Code 5 Lead.
Hot Flues: To hot flues/exhaust pipes install a galvanised steel sleeve, allowing a **50mm** void between the sleeve and the pipe, which is to be filled with non-combustible insulation. A galvanised steel cowling is to be installed to cover.

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CONTRACTORS / INSURANCE BACKED GUARANTEE

Photographic Evidence: The Contractor is to record with photographic evidence the surface preparation, priming and each layer installed, standard detailing and terminations, non-standard detailing, and terminations to be approved by **PROTEUS WATERPROOFING**.
Inspection: The roof may be inspected by **PROTEUS WATERPROOFING,** and progress reports undertaken to review the work being carried out as per the Proteus Specification. The works may benefit from an inspection to review the general competence of operatives and standard of workmanship. **Safe access must be available for any inspection to be carried out.**
Proteus Snag List: A snagging list may be produced to address any areas noted that are sub-standard or not conforming with the requirements of the specification.
Final Sign off: At the completion of the specified project, works may be signed off by **PROTEUS WATERPROOFING** to generally review the scope of the Proteus Specification and the final Finished System.
Contractors: **PROTEUS WATERPROOFING** approved Contractors must have been assessed in the specified membrane. The Contractor must offer reasonable notice to **PROTEUS WATERPROOFING** of commencement of the works if any inspections are to be scheduled throughout the works program. The Guarantee can be issued following a final inspection of the works, as carried out by the Contractor. The Contractor is to request any final inspection on practical completion of the contract. All snagging and corrections must be made prior the final inspection.
Proteus Insurance Backed Guarantee: A **20 Year Proteus Insurance Backed Guarantee** is required providing **20 years** of cover with insurance backing for the first **12 years**; subject to the current terms and conditions of the guarantee which are available separately upon request. The installation of the Finished System within the Proteus Specification shall be carried out by a competent Contractor. Note: The insurance is an independent pre-paid policy supplied by an FCA regulated provider.

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